MATH 243 — Probability and Statistics

Correlation and Regression Analysis

Lesson notes · Exam approach · TF practice with solutions · Quick notes (Statistics portion)

How to Read These Notes

EXAM FOCUS	Exam importance, where it appears in the TF collection
MEMORIZE	Rules, tables and formulas to memorize
TRAP	Common mistakes and exam traps
EXAM-READY ANSWER	Model wording you can write in the exam
NOTE	Extra explanation / clarification
Grey box, monospace	Code, worked tables, derivations and hand-drawn sketches
Centered grey box	Formula to remember / final numerical result
Purple heading / ✓ green	Practice questions / their solutions

Contents

Exam Map — What the TF Papers Ask — a regression question in every paper
Lesson 12 — Correlation, Scatter Diagram and Pearson's r — types, interpretation, $-1 \le r \le 1$
Lesson 13 — Spearman's Rank Correlation (with Ties) — D^2, tie correction
Lesson 14 — The Two Regression Lines — least squares, b_{yx}, b_{xy}, estimates, all TF tables
Lesson 15 — Goodness of Fit (R^2), Polynomial and Multiple Regression — SST = SSR + SSE, quadratic fit
Quick Notes — Correlation and Regression — revision bullets

NOTE Page numbers refer to 7. Correlation and regression analysis.pdf (17 pp.), written as p. n. Notation: $S_{xx} = \Sigma x^2 - (\Sigma x)^2/n$, $S_{yy} = \Sigma y^2 - (\Sigma y)^2/n$, $S_{xy} = \Sigma xy - \Sigma x \Sigma y/n$. "TF p." = merged question PDF page; "22 batch p." = 2023-24 paper.

Exam Map — What the TF Papers Ask

Session (paper)	Questions answered by this note [lesson]
2023-24 (MATH 243) TF p. 22 batch p.2, 22 batch p.3	Q4(b) Pearson's r; rank correlation with repeated ranks — sales and expenses of 10 stores (12) [L13] · Q4(c) Objectives of correlation/regression; second-degree polynomial for nitrogen vs yield; coefficient of determination (12) [L15] · Q5(b) Scatter plot of price vs rating for 5 restaurants; is there a relationship? (13) [L12]
2022-23 (MATH 243) TF p. 48, 49	Q4(b) Correlation vs regression; rank correlation, Bangla vs English marks of 8 applicants (ties) (12) [L13] · Q4(c) Ages of 10 husbands and wives: two regression equations; husband at wife 30, wife at husband 25; comment on r (13) [L14]
2021-22 (MATH 245) TF p. 44–45	Q7(a) Loblolly pines (15): specific gravity vs modulus of rupture — both regression coefficients; r and interpretation (20) [L14]
2020-21 (MATH 245) TF p. 37	Q6(a) Experience X vs rating Y (5 persons): fit Y on X; SSR and SST (15) [L15]
2019-20 (MATH 245) TF p. 34	Q6(a) Distance to transmitter vs signal strength (10 pairs): regression of Y on X; predict at 10 m (17) [L14]
2018-19 (MATH 245) TF p. 24	Q6(a) Regression of capacity utilisation on production from means, SDs and r = 0.62; production when capacity is 70 (20) [L14]
2016-17 (MATH 245) TF p. 20	Q6(b) Experience X vs rating Y (5 persons): fit Y on X; SSR, SST; is the linear model adequate (R^2)? (18) [L15]
2015-16 (MATH 241) TF p. 16	Q7(b) Years of service vs income (9 persons): regression equations, r; starting income for 15 years' service (17) [L14]
2014-15 (MATH 241) TF p. 11	Q7(a) Regression lines Y on X and X on Y and r (7 pairs, X 42…66, Y 56…51) (20) [L14]
2013-14 (MATH 241) TF p. 8	Q7(a) Ages of husbands and wives (8 couples): two regression equations, r; husband's age when wife is 24 (20) [L14]
2012-13 (MATH 241) TF p. 5	Q7(a) Economics vs Business marks (10 students): two regression equations and coefficients; estimates; comment on r (18) [L14]
2011-12 (MATH 241) TF p. 2	Q7(a) Two regression lines and r from 9 (X, Y) pairs (X 105…111, Y 58…55) (20) [L14]

EXAM FOCUS 15 questions, 247 marks — the largest Statistics topic, in every paper. Nine of them are the same task: from paired data find both regression lines, r and an estimate. Since 2016-17 the papers add SSR/SST/R^2 adequacy, rank correlation with ties (2022-23, 2023-24) and a second-degree polynomial (2023-24). Learn the S_{xx}, S_{yy}, S_{xy} table — it gives every answer.

Lesson 12 — Correlation, Scatter Diagram and Pearson's r

pp. 1–5

Correlation analysis measures the strength (degree) of the relationship between variables (fertiliser and rice production, height and weight). Positive: both increase together; negative: one increases as the other decreases; perfect: the ratio of changes is constant (p. 1). Types: positive/negative, simple/multiple, linear/non-linear.

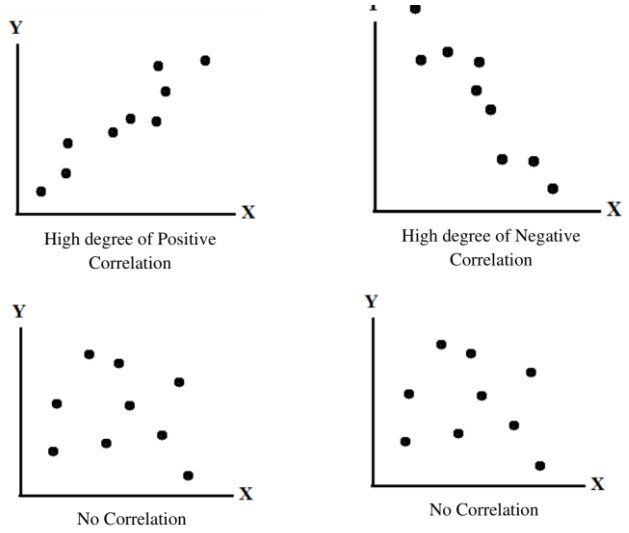


Figure 12.1 — Scatter diagrams: high positive, high negative and no correlation (p. 2)

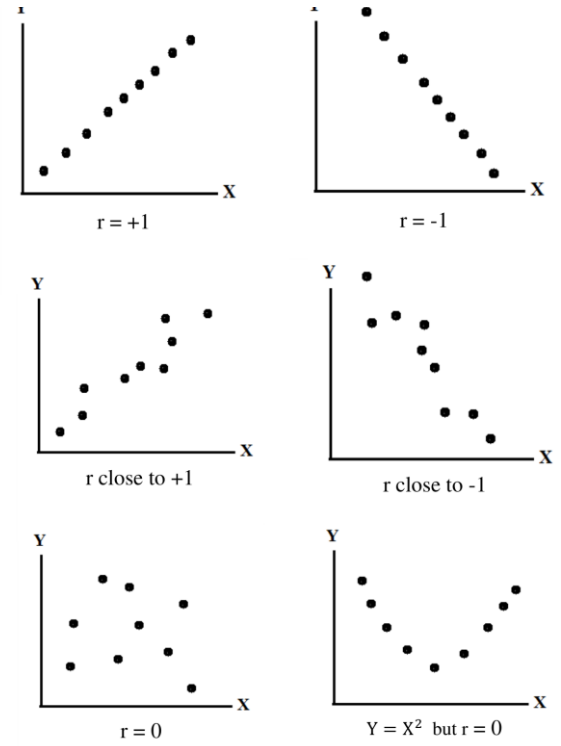


Figure 12.2 — r = ±1, r close to ±1, r = 0, and Y = X² with r = 0 (p. 3)

$$r = \Sigma(x - \bar{x})(y - \bar{y}) \,/\, \sqrt{[\Sigma(x - \bar{x})^2 \, \Sigma(y - \bar{y})^2]} = S_{xy}/\sqrt{(S_{xx}S_{yy})}$$

- $-1 \le r \le 1$. $r = +1$ / -1: perfect positive / negative linear relation.
- r near ±1: strong linear relation; near 0: weak linear relation. r = 0 does not mean "no relation" — the relation may be curved (p. 3).

Proof that −1 ≤ r ≤ 1 (p. 5)

```
Let u = (x − x̄)/√Σ(x − x̄)²,  v = (y − ȳ)/√Σ(y − ȳ)²,  so Σu² = Σv² = 1 and Σuv = r.
0 ≤ Σ(u + v)² = Σu² + Σv² + 2Σuv = 2 + 2r   ⟹  r ≥ −1
0 ≤ Σ(u − v)² = 2 − 2r                      ⟹  r ≤ 1
```

Sales vs expenses of 10 firms (p. 4): $\Sigma(x - \bar{x})^2 = 360$, $\Sigma(y - \bar{y})^2 = 22$, $\Sigma(x - \bar{x})(y - \bar{y}) = 70$ → $r = 70/\sqrt{(360 \times 22)} = 0.7866$, a fairly high positive correlation. Advertising vs sales: r = 0.865.

Practice — Lesson 12

(a) Actual — 2023-24 Q5(b) (22 batch p. 3) — the plot is drawn in the Intro & Graphs note, Lesson 3.

(a) Prices ($) and ratings of 5 restaurants: (22, 3.5), (29, 4.0), (15, 3.0), (18, 4.0), (23, 3.5). Draw the scatter plot and interpret whether price and rating are related. (13)

✓ Solution

```
Σx = 107, Σy = 18, Σx² = 2403, Σy² = 65.5, Σxy = 390.5
S_xx = 2403 − 107²/5 = 113.2;  S_yy = 65.5 − 18²/5 = 0.7;  S_xy = 390.5 − 107×18/5 = 5.3
r = 5.3/√(113.2 × 0.7) = 0.595
```

The points drift upward with one exception (D): a moderate positive relationship (r ≈ 0.60) — higher prices tend to go with higher ratings, but with five restaurants the evidence is weak.

Lesson 13 — Spearman's Rank Correlation (with Ties)

pp. 5–7

Used when only ranks are available or the characteristic cannot be measured exactly (sincerity, honesty).

$$r_s = 1 - 6\Sigma D^2 / [n(n^2 - 1)]$$
$$\text{Ties: } r_s = 1 - 6[\Sigma D^2 + \Sigma(m^3 - m)/12] / [n(n^2 - 1)]$$

D = difference between the two ranks; tied values get the average of the ranks they occupy; m = number of times a value is repeated (one correction term per tied group, in either series). Managers: $\Sigma D^2 = 14 \rightarrow r_s = 1 - 84/990 = 0.915$ (high agreement). Tests: $\Sigma D^2 = 44 \rightarrow r_s = 0.733$.

Exam Approach

1. Rank each series separately (largest = highest rank or smallest = 1 — same convention for both).
2. Average ranks for ties; list D and D^2; add the $(m^3 - m)/12$ corrections.
3. Substitute; interpret sign and size.

Practice — Lesson 13

(a) Actual — 2022-23 Q4(b) (TF p. 48). (b) Actual — 2023-24 Q4(b) (22 batch p. 2).

(a) Distinguish clearly between correlation and regression analysis. Marks of 8 applicants — Bangla: 15, 20, 28, 12, 40, 60, 20, 80; English: 40, 30, 50, 30, 20, 10, 30, 60. Compute the rank correlation coefficient and comment. (12)

(b) What is Pearson's correlation coefficient? How do you modify the rank formula when observations are repeated? Hourly sales x: 50, 50, 55, 60, 65, 65, 65, 60, 60, 60 and expenses y: 11, 13, 14, 16, 16, 15, 15, 14, 13, 13 of 10 stores: calculate the rank correlation coefficient. (12)

✓ Solution

EXAM-READY ANSWER Correlation vs regression (p. 14): correlation measures the degree and direction of association and treats X and Y symmetrically ($r_{xy} = r_{yx}$, unit-free, −1 to 1); regression expresses the average dependence of a dependent variable on independent variable(s) through an equation, treats them asymmetrically ($b_{yx} \neq b_{xy}$, in units of the data) and is used to predict. Correlation does not imply causation; regression assumes a direction of dependence.

(a) Ranks (1 = smallest).

Applicant	Bangla	R_B	English	R_E	D	D^2
A	15	2	40	6	−4	16
B	20	3.5	30	4	−0.5	0.25
C	28	5	50	7	−2	4
D	12	1	30	4	−3	9
E	40	6	20	2	4	16
F	60	7	10	1	6	36
G	20	3.5	30	4	−0.5	0.25
H	80	8	60	8	0	0
Total					0	81.5

```
Ties: Bangla 20 twice (m = 2): (2³ − 2)/12 = 0.5;  English 30 three times (m = 3): (3³ − 3)/12
= 2
r_s = 1 − 6(81.5 + 0.5 + 2)/[8(64 − 1)] = 1 − 504/504 = 0
```

$r_s = 0$: no association between the rankings in Bangla and English. (The slide on p. 7 uses the same data.)

(b) Pearson's r: §L12. Modified formula: add $\Sigma(m^3 - m)/12$ to ΣD^2 for every tied group.

x	R_x	y	R_y	D²
50	1.5	11	1	0.25
50	1.5	13	3	2.25
55	3	14	5.5	6.25
60	5.5	16	9.5	16
65	9	16	9.5	0.25
65	9	15	7.5	2.25
65	9	15	7.5	2.25
60	5.5	14	5.5	0
60	5.5	13	3	6.25
60	5.5	13	3	6.25

```
ΣD² = 42. Ties in x: 50 (m = 2), 60 (m = 4), 65 (m = 3); in y: 13 (m = 3), 14 (m = 2), 15 (m =
2), 16 (m = 2)
Σ(m³ − m)/12 = 0.5 + 5 + 2 + 2 + 0.5 + 0.5 + 0.5 = 11
r_s = 1 − 6(42 + 11)/[10(100 − 1)] = 0.6788
```

A fairly high positive rank correlation (0.68): stores with higher sales tend to have higher expenses.

NOTE The paper prints the last sales value as 60; the slide (p. 4) has 50 for firm 10. With the slide's data r_s = 0.7515 and Pearson r = 0.787. Use the paper's numbers in the exam.

Lesson 14 — The Two Regression Lines

pp. 8–12

Regression studies the dependence of a dependent variable on independent variable(s), to estimate or predict its average value. Objectives (p. 9): estimate the average relationship; find the effect of each independent variable; predict the dependent variable. A deterministic relation is exact; a non-deterministic one has random error ε.

$$\textbf{Y on X: } \hat{y} = a + b_{yx}x,\ b_{yx} = [n\Sigma xy - \Sigma x\Sigma y]/[n\Sigma x^2 - (\Sigma x)^2] = S_{xy}/S_{xx},\ a = \bar{y} - b_{yx}\bar{x}$$

$$\textbf{X on Y: } \hat{x} = a' + b_{xy}y,\ b_{xy} = S_{xy}/S_{yy},\ a' = \bar{x} - b_{xy}\bar{y}$$

$$r = \pm\sqrt{(b_{yx}b_{xy})};\ b_{yx} = r\,\sigma_y/\sigma_x,\ b_{xy} = r\,\sigma_x/\sigma_y$$

Least squares (p. 10): choose a, b to minimise SSE = $\Sigma(y - a - bx)^2$; $\partial SSE/\partial a = 0$ and $\partial SSE/\partial b = 0$ give the normal equations $\Sigma y = na + b\Sigma x$ and $\Sigma xy = a\Sigma x + b\Sigma x^2$.

Transmitter example (p. 11): $\Sigma x = 118$, $\Sigma y = 336.6$, $\Sigma x^2 = 1708$, $\Sigma xy = 3834.3$ → $b = (10\times3834.3 - 118\times336.6)/(10\times1708 - 118^2) = -0.4359$, $a = 38.804$; $\hat{y} = 38.804 - 0.4359x$; at 10 m $\hat{y} = 34.44$ dB.

MEMORIZE Use the line of the variable you want to estimate: estimate y from x with Y on X; estimate x from y with X on Y. Both lines pass through $(\bar{x}, \bar{y})$; r has the sign of the b's.

Exam Approach

1. Table of x, y, x^2, y^2, xy (or deviations); totals; $\bar{x}$, $\bar{y}$.
2. S_{xx}, S_{yy}, S_{xy}; $b_{yx} = S_{xy}/S_{xx}$, $b_{xy} = S_{xy}/S_{yy}$; intercepts from the means.
3. $r = S_{xy}/\sqrt{(S_{xx}\,S_{yy})}$ (check $r^2 = b_{yx}\,b_{xy}$); interpret.
4. Estimates from the correct line; state units.

Practice — Lesson 14

(a) Actual — 2022-23 Q4(c) (TF p. 49). (b) Actual — 2019-20 Q6(a) (TF p. 34). (c) Actual — 2018-19 Q6(a) (TF p. 24). (d) Actual — 2021-22 Q7(a) (TF pp. 44–45). (e) Actual — 2015-16 Q7(b) (TF p. 16). (f) Actual — 2014-15 Q7(a) (TF p. 11). (g) Actual — 2013-14 Q7(a) (TF p. 8). (h) Actual — 2012-13 Q7(a) (TF p. 5). (i) Actual — 2011-12 Q7(a) (TF p. 2).

(a) Ages of husbands 22, 23, 23, 24, 26, 27, 27, 28, 30, 30 and wives 18, 20, 21, 20, 21, 22, 23, 24, 25, 26: the two regression equations; husband's age when the wife is 30; wife's age when the husband is 25; comment on r. (13)

(b) Distance to transmitter X: 13, 12, 17, 19, 14, 15, 15, 8, 13, 3 and signal strength Y: 34.4, 38.4, 30.4, 29.7, 30.1, 33.9, 32.8, 35.2, 34.9, 36.8. Regression line of Y on X; predict Y at 10 m. (17)

(c) Production: mean 35.6, SD 10.5; capacity utilisation: mean 84.8, SD 8.5; r = 0.62. Regression of capacity utilisation on production; estimate production when capacity utilisation is 70 %. (20)

(d) Specific gravity x and modulus of rupture y of 15 loblolly pines (TF p. 45): both regression coefficients; compute and interpret r. (20)

(e) Years of service 10, 8, 9, 5, 7, 6, 12, 11, 13 and income ('000) 17, 12, 14, 8, 10, 9, 20, 18, 22: regression equations and r; starting income for 15 years' experience. (17)

(f) X: 42, 44, 58, 55, 89, 98, 66; Y: 56, 49, 53, 58, 65, 76, 51 — regression lines Y on X and X on Y; r. (20)

(g) Husbands 22, 25, 28, 31, 35, 32, 37, 39; wives 19, 18, 22, 29, 31, 23, 30, 33: regression equations, r; most likely age of husband when the wife is 24. (20)

(h) Economics 47, 38, 34, 44, 32, 41, 35, 32, 42, 30; Business 38, 34, 42, 39, 34, 33, 30, 41, 38, 28: regression equations and coefficients; Economics when Business = 36; Business when Economics = 44; comment on r. (18)

(i) X: 105, 112, 104, 114, 117, 108, 106, 117, 111; Y: 58, 56, 47, 46, 51, 64, 66, 50, 55 — two regression lines, r. (20)

✓ Solution

(a) Let x = husband's age, y = wife's age; n = 10.

```
Σx = 260, Σy = 220, Σx² = 6836, Σy² = 4896, Σxy = 5782;  x̄ = 26, ȳ = 22
S_xx = 76,  S_yy = 56,  S_xy = 62
b_yx = 62/76 = 0.8158;   Y on X: ŷ = 0.7895 + (0.8158)x
b_xy = 62/56 = 1.1071;   X on Y: x̂ = 1.6429 + (1.1071)y
r = 62/√(76 × 56) = 0.9504   (check √(b_yx b_xy) = 0.9504)
Husband at wife = 30: x̂ = 1.6429 + 1.1071(30) = 34.86 years
Wife at husband = 25: ŷ = 0.7895 + 0.8158(25) = 21.18 years
```

r = 0.95: very high positive correlation — older husbands have older wives.

(b) Let x = distance (m), y = signal (dB); n = 10.

```
Σx = 129, Σy = 336.6, Σx² = 1851, Σy² = 11406.52, Σxy = 4256.7;  x̄ = 12.9, ȳ = 33.66
S_xx = 186.9,  S_yy = 76.564,  S_xy = −85.44
b_yx = −85.44/186.9 = −0.4571;   Y on X: ŷ = 39.5571 + (−0.4571)x
b_xy = −85.44/76.564 = −1.1159;   X on Y: x̂ = 50.4622 + (−1.1159)y
r = −85.44/√(186.9 × 76.564) = −0.7142   (check √(b_yx b_xy) = 0.7142)
At x = 10: ŷ = 39.5571 − 0.4571(10) = 34.99 dB
```

Signal strength falls about 0.46 dB per metre. (With the slide's value 1 instead of 12 for the second distance the line is ŷ = 38.804 − 0.4359x and the prediction 34.44 dB.)

(c) x = production, y = capacity utilisation.

```
b_yx = r σ_y/σ_x = 0.62 × 8.5/10.5 = 0.5019;  y − 84.8 = 0.5019(x − 35.6)  →  ŷ = 66.9322 +
0.5019x
b_xy = r σ_x/σ_y = 0.62 × 10.5/8.5 = 0.7659;  x − 35.6 = 0.7659(y − 84.8)  →  x̂ = −29.3468 +
0.7659y
Production when y = 70: x̂ = 35.6 + 0.7659(70 − 84.8) = 24.26
```

(d) Let x = specific gravity (g/cm³), y = modulus of rupture (kPa); n = 15.

```
Σx = 7.296, Σy = 788,510, Σx² = 3.6201, Σy² = 47,598,707,772, Σxy = 403686.826;  x̄ = 0.4864, ȳ
= 52567.3333
S_xx = 0.0714,  S_yy = 6148839765.3333,  S_xy = 20155.562
b_yx = 20155.562/0.0714 = 282458.5188;   Y on X: ŷ = −84820.4902 + (282458.5188)x
b_xy = 20155.562/6148839765.3333 = 0;   X on Y: x̂ = 0.3141 + (0)y
r = 20155.562/√(0.0714 × 6148839765.3333) = 0.9622   (check √(b_yx b_xy) = 0.9622)
```

b_yx ≈ 282,459 kPa per g/cm³ (an increase of 0.01 in specific gravity raises the modulus by about 2,825 kPa); b_xy ≈ 3.28 × 10^{-6} g/cm³ per kPa. r = 0.962: a very strong positive linear relationship — denser wood is stronger; r^2 = 0.926, so 92.6 % of the variation in modulus of rupture is explained by specific gravity.

(e) Let x = years of service, y = income ('000); n = 9.

```
Σx = 81, Σy = 130, Σx² = 789, Σy² = 2082, Σxy = 1280;  x̄ = 9, ȳ = 14.4444
S_xx = 60,  S_yy = 204.2222,  S_xy = 110
b_yx = 110/60 = 1.8333;   Y on X: ŷ = −2.0556 + (1.8333)x
b_xy = 110/204.2222 = 0.5386;   X on Y: x̂ = 1.2198 + (0.5386)y
r = 110/√(60 × 204.2222) = 0.9937   (check √(b_yx b_xy) = 0.9937)
At x = 15: ŷ = −2.0556 + 1.8333(15) = 25.44 thousand
```

r = 0.994: almost perfect positive correlation; recommend a starting income of about Tk 25,440.

(f) Let x = X, y = Y; n = 7.

```
Σx = 452, Σy = 408, Σx² = 31,970, Σy² = 24,312, Σxy = 27,371;  x̄ = 64.5714, ȳ = 58.2857
S_xx = 2783.7143,  S_yy = 531.4286,  S_xy = 1025.8571
b_yx = 1025.8571/2783.7143 = 0.3685;   Y on X: ŷ = 34.4898 + (0.3685)x
b_xy = 1025.8571/531.4286 = 1.9304;   X on Y: x̂ = −47.9419 + (1.9304)y
r = 1025.8571/√(2783.7143 × 531.4286) = 0.8434   (check √(b_yx b_xy) = 0.8434)
```

r = 0.84: high positive correlation.

(g) Let x = husband's age, y = wife's age; n = 8.

```
Σx = 249, Σy = 205, Σx² = 7993, Σy² = 5489, Σxy = 6601;  x̄ = 31.125, ȳ = 25.625
S_xx = 242.875,  S_yy = 235.875,  S_xy = 220.375
b_yx = 220.375/242.875 = 0.9074;   Y on X: ŷ = −2.6166 + (0.9074)x
b_xy = 220.375/235.875 = 0.9343;   X on Y: x̂ = 7.1839 + (0.9343)y
r = 220.375/√(242.875 × 235.875) = 0.9207   (check √(b_yx b_xy) = 0.9207)
Husband at wife = 24: x̂ = 7.1839 + 0.9343(24) = 29.61 years
```

r = 0.92: high positive correlation.

(h) Let x = Economics, y = Business; n = 10.

```
Σx = 375, Σy = 357, Σx² = 14,363, Σy² = 12,939, Σxy = 13,461;  x̄ = 37.5, ȳ = 35.7
S_xx = 300.5,  S_yy = 194.1,  S_xy = 73.5
b_yx = 73.5/300.5 = 0.2446;   Y on X: ŷ = 26.5278 + (0.2446)x
b_xy = 73.5/194.1 = 0.3787;   X on Y: x̂ = 23.9815 + (0.3787)y
r = 73.5/√(300.5 × 194.1) = 0.3043   (check √(b_yx b_xy) = 0.3043)
Economics at Business = 36: x̂ = 23.9815 + 0.3787(36) = 37.61
Business at Economics = 44: ŷ = 26.5278 + 0.2446(44) = 37.29
```

Regression coefficients b_yx = 0.245, b_xy = 0.379; r = 0.30 — a weak positive correlation, so the estimates are not reliable.

(i) Let x = X, y = Y; n = 9.

```
Σx = 994, Σy = 493, Σx² = 109,980, Σy² = 27,403, Σxy = 54,324;  x̄ = 110.4444, ȳ = 54.7778
S_xx = 198.2222,  S_yy = 397.5556,  S_xy = −125.1111
b_yx = −125.1111/198.2222 = −0.6312;   Y on X: ŷ = 124.4865 + (−0.6312)x
b_xy = −125.1111/397.5556 = −0.3147;   X on Y: x̂ = 127.6831 + (−0.3147)y
r = −125.1111/√(198.2222 × 397.5556) = −0.4457   (check √(b_yx b_xy) = 0.4457)
```

r = −0.45: moderate negative correlation.

Lesson 15 — Goodness of Fit (R^2), Polynomial and Multiple Regression

pp. 12–17

$$\mathbf{SST = \Sigma(y - \bar{y})^2 = SSR + SSE,\quad SSR = \Sigma(\hat{y} - \bar{y})^2,\quad SSE = \Sigma(y - \hat{y})^2}$$
$$\mathbf{R^2 = SSR/SST,\ 0 \le R^2 \le 1;\ \text{simple linear: } R^2 = r^2,\ SSR = b_{yx}S_{xy}}$$

R^2 is the proportion (percentage) of the total variation in y explained by the regression. Close to 1 → the model fits well; deterministic relation → SSE = 0, R^2 = 1. Hardness vs tensile strength (p. 14): $\hat{y}$ = 14.37 + 0.4167x, R^2 = 0.51 — only half the variation explained, so the linear model is not very adequate.

Multiple and Polynomial Regression (pp. 14–17)

$$\mathbf{\hat{y} = b_0 + b_1x_1 + b_2x_2:\ \Sigma y = nb_0 + b_1\Sigma x_1 + b_2\Sigma x_2,\ \Sigma x_1y = b_0\Sigma x_1 + b_1\Sigma x_1^2 + b_2\Sigma x_1x_2,\ \Sigma x_2y = \ldots}$$
$$\mathbf{\hat{y} = a + bx + cx^2\text{: treat x as } x_1 \text{ and } x^2 \text{ as } x_2}$$

Rice (p. 15): rainfall x_1, urea x_2: least squares gives $\hat{y} = 27.355 - 0.6295x_1 - 0.0601x_2$.

Exam Approach

1. Fit the line; $\hat{y}$ for each x; SSR = $\Sigma(\hat{y} - \bar{y})^2$ (or b S_{xy}), SST = S_{yy}, SSE = SST − SSR.
2. R^2 = SSR/SST; ≥ 0.8 adequate, around 0.5 or below doubtful.
3. Quadratic: compute Σx, Σx^2, Σx^3, Σx^4, Σy, Σxy, Σx^2y; solve the three normal equations; $R^2 = 1 - SSE/SST$.

Practice — Lesson 15

(a) Actual — 2016-17 Q6(b) (TF p. 20) = 2020-21 Q6(a) (TF p. 37). (b) Actual — 2023-24 Q4(c) (22 batch p. 2).

(a) Experience X: 16, 12, 18, 4, 3 and performance rating Y: 87, 88, 89, 68, 58. (i) Fit a linear regression of Y on X. (ii) Find SSR and SST. (iii) Is the linear model appropriate? Justify with the coefficient of determination. (18)

(b) Briefly mention the objectives of correlation and regression analysis. Nitrogen doses x: 0, 1, 2, 3, 4, 5 (lb/acre) and yield y: 15, 25, 40, 55, 52, 43 (maunds/acre). Fit a second-degree polynomial and obtain the coefficient of determination. (12)

✓ Solution

```
Σx = 53, Σy = 390, Σx² = 749, Σxy = 4496, Σy² = 31222;  x̄ = 10.6, ȳ = 78
S_xx = 749 − 53²/5 = 187.2;  S_xy = 4496 − 53×390/5 = 362;  S_yy = 31222 − 390²/5 = 802
(i)   b = 362/187.2 = 1.9338;  a = 78 − 1.9338 × 10.6 = 57.5021;  ŷ = 57.5021 + 1.9338x
      fitted ŷ: 88.44, 80.71, 92.31, 65.24, 63.3
(ii)  SSR = Σ(ŷ − ȳ)² = b S_xy = 1.9338 × 362 = 700.02;  SST = S_yy = 802
(iii) R² = SSR/SST = 700.02/802 = 0.8728
```

About 87 % of the variation in performance rating is explained by experience, so the linear model is appropriate (SSE = 101.98 is small relative to SST).

(b) Objectives: L14 (estimate the average relationship, measure the effect of each independent variable, predict). Normal equations for $\hat{y} = a + bx + cx^2$ (n = 6):

```
Σx = 15, Σx² = 55, Σx³ = 225, Σx⁴ = 979, Σy = 230, Σxy = 693, Σx²y = 2587
230  = 6a  + 15b  + 55c
693  = 15a + 55b  + 225c
2587 = 55a + 225b + 979c
Solving: a = 11.5357,  b = 21.6536,  c = −2.9821
ŷ = 11.536 + 21.654x − 2.982x²
Fitted: 11.54, 30.21, 42.91, 49.66, 50.44, 45.25
SST = Σ(y − ȳ)² = 1211.33;  SSE = Σ(y − ŷ)² = 83.66;  R² = 1 − SSE/SST = 0.9309
```

$R^2 = 0.931$: the quadratic explains 93 % of the variation in yield. Yield is maximum near $x = -b/(2c) = 3.63$ lb/acre.

NOTE The slide (p. 17) uses the same data; its working table matches these sums.

Quick Notes — Correlation and Regression

Revision points for Lessons 12–15.

L12 · Pearson r

- $r = S_{xy}/\sqrt{(S_{xx}\, S_{yy})}$, $-1 \le r \le 1$ (proof with $\Sigma(u \pm v)^2 \ge 0$); $r = 0 \neq$ no relation.
- Sales/expenses $r = 0.787$; restaurants $r = 0.60$.

L13 · Rank Correlation

- $r_s = 1 - 6[\Sigma D^2 + \Sigma(m^3 - m)/12]/[n(n^2 - 1)]$.
- Bangla/English: $\Sigma D^2 = 81.5$, correction $2.5 \rightarrow r_s = 0$. Stores: $r_s = 0.679$ (paper data; 0.752 with the slide data).

L14 · Regression Lines

- $b_{yx} = S_{xy}/S_{xx}$, $b_{xy} = S_{xy}/S_{yy}$, lines through $(\bar{x}, \bar{y})$; $r^2 = b_{yx}\, b_{xy}$.
- From summary data: $b_{yx} = r\sigma_y/\sigma_x$ (capacity: $\hat{y} = 66.93 + 0.502x$; production at 70: 24.26).
- Husband/wife 2022-23: $r = 0.95$, 34.86 and 21.18. Service/income: 25.44. Transmitter (paper): 34.99 dB.

L15 · R^2 and Polynomial

- $SST = SSR + SSE$; $R^2 = SSR/SST = r^2$ for a line. Experience/rating $R^2 = 0.873$.
- Quadratic: three normal equations; nitrogen $\hat{y} = 11.536 + 21.654x - 2.982x^2$, $R^2 = 0.931$.